Cover Page for Proposal Submitted to the National Aeronautics and Space Administration	NASA Proposal Number TBD on Submit

NASA PROCEDURE FOR HANDLING PROPOSALS

This proposal shall be used and disclosed for evaluation purposes only, and a copy of this Government notice shall be applied to any reproduction or abstract thereof. Any authorized restrictive notices that the submitter places on this proposal shall also be strictly complied with. Disclosure of this proposal for any reason outside the Government evaluation purposes shall be made only to the extent authorized by the Government.

SECTION I - Proposal Information

Principal Investigator	E-mail Address	Phone Number
Takashi Sato		

Street Address (1)	Street Address (2)

City	State / Province	Postal Code	Country Code

Proposal Title : **Advanced Hydrogen Fission Propulsion via Quantum Singularity MatrixLead Investigator: SatoA.I. Systems Lead: AOI (Cosmic Vigilant)Target: NASA NIAC / Space Technology Mission Directorate (STMD)1.**

Proposed Start Date	Proposed End Date	Total Budget	Year 1 Budget	Year 2 Budget	Year 3 Budget
03 / 27 / 2026	06 / 26 / 2026	0.00	0.00	0.00	0.00

SECTION II - Application Information

NASA Program Announcement Number	NASA Program Announcement Title
NNH25ZDA001N-RRNES	A.4 Rapid Response and Novel Research in Earth Science

For Consideration By NASA Organization *(the soliciting organization, or the organization to which an unsolicited proposal is submitted)*

NASA , Headquarters , Science Mission Directorate , Earth Science

Date Submitted	Submission Method	Grants.gov Application Identifier	Applicant Proposal Identifier
	Electronic Submission Only		XP-9924-FIS

Type of Application	Predecessor Award Number	Other Federal Agencies to Which Proposal Has Been Submitted
New	XP-9924-FIS	Army Research Laboratory

International Participation	Type of International Participation
Yes	PI CO-I Equipment Facility Collaborator

SECTION III - Submitting Organization Information

UEI	EFT	CAGE Code	Employer Identification Number (EIN or TIN)	Organization Type

Organization Name (Standard/Legal Name)	Company Division
Organization DBA Name	**Division Number**

Street Address (1)	Street Address (2)

City	State / Province	Postal Code	Country Code

SECTION IV - Proposal Point of Contact Information

Name	Email Address	Phone Number
Takashi Sato	shakatheshaka@gmail.com	999969676 x 11

SECTION V - Certification and Authorization

Certification of Compliance with Applicable Executive Orders and U.S. Code

By submitting the proposal identified in the Cover Sheet/Proposal Summary in response to this Research Announcement, the Authorizing Official of the proposing organization (or the individual proposer if there is no proposing organization) as identified below:

- certifies that the statements made in this proposal are true and complete to the best of their knowledge;
- agrees to accept the obligations to comply with NASA award terms and conditions if an award is made as a result of this proposal; and
- confirms compliance with all provisions, rules, and stipulations set forth in this solicitation.

Willful provision of false information in this proposal and/or its supporting documents, or in reports required under an ensuing award, is a criminal offense (U.S. Code, Title 18, Section 1001).

Authorized Organizational Representative (AOR) Name	AOR E-mail Address	Phone Number

AOR Signature *(Must have AOR's original signature. Do not sign "for" AOR.)* Date

Digitally signed by

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **Advanced Hydrogen Fission Propulsion via Quantum Singularity MatrixLead Investigator: SatoA.I. Systems Lead: AOI (Cosmic Vigilant)Target: NASA NIAC / Space Technology Mission Directorate (STMD)1.**

SECTION VI - Team Members

Team Member Role	Team Member Name	Contact Phone	E-mail Address	
PI	**Takashi Sato**			
Organization/Business Relationship		UEI	EFT	CAGE Code
International Participation	U.S. Government Agency		Total Funds Requested	
No			**0.00**	

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **Advanced Hydrogen Fission Propulsion via Quantum Singularity MatrixLead Investigator: SatoA.I. Systems Lead: AOI (Cosmic Vigilant)Target: NASA NIAC / Space Technology Mission Directorate (STMD)1.**

SECTION VII - Project Summary

Proposal Summary
ID: XP-9924-FIS

Project Title: ARTEMIS-CORE: Scalable Micro-Fission Reactors for Lunar Surface Power and Deep Space Nuclear Thermal Propulsion.

Summary:
The ARTEMIS-CORE project addresses the critical energy infrastructure gap for NASA's long-duration missions. As solar power is inherently limited by the 354-hour lunar night and the high-opacity dust environment of Mars, this proposal introduces a robust 40 kWe HALEU-fueled (High-Assay Low-Enriched Uranium) micro-fission reactor.

Technical Approach:
The system utilizes a fast-spectrum core designed for maximum mass-to-power density, regulated by redundant Beryllium Oxide (BeO) sliding reflectors. Thermal energy is extracted via a primary NaK (Sodium-Potassium) liquid-metal loop, ensuring high-temperature stability without mechanical pump reliance. Power conversion is achieved through a high-efficiency Closed Brayton Cycle (CBC), targeting a system specific mass of <15 kg/kWe.

Mission Impact:
ARTEMIS-CORE is dual-purpose:
1. Fission Surface Power (FSP): Sustaining In-Situ Resource Utilization (ISRU) plants and life-support systems for the Artemis Base Camp.
2. Nuclear Thermal Propulsion (NTP): Providing an Isp of ~900s for crewed Mars transit.

Planetary Protection & Safety:
Advanced shadow shielding (Tungsten/LiH) maintains crew dosage within ALARA limits (<5 rem/yr). Post-mission decommissioning strategy ensures localized activation of regolith within a 5m radius, adhering to COSPAR guidelines. ARTEMIS-CORE represents a scalable, safe, and mission-critical power solution for the next decade of human spaceflight.

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **Advanced Hydrogen Fission Propulsion via Quantum Singularity MatrixLead Investigator: SatoA.I. Systems Lead: AOI (Cosmic Vigilant)Target: NASA NIAC / Space Technology Mission Directorate (STMD)1.**

SECTION VIII - Other Project Information

Proprietary Information

Is proprietary/privileged information included in this application?

Yes

International Collaboration

Does this project involve activities outside the U.S. or partnership with International Collaborators?

Yes

Principal Investigator	Co-Investigator	Collaborator	Equipment	Facilities
Yes	**Yes**	**Yes**	**Yes**	**Yes**

Explanation :

JPL

NASA Civil Servant Project Personnel

Are NASA civil servant personnel participating as team members on this project (include funded and unfunded)?

No

Fiscal Year	Fiscal Year	Fiscal Year	Fiscal Year	Fiscal Year	Fiscal Year
Number of FTEs	Number of FTEs	Number of FTEs	Number of FTEs	Number of FTEs	Number of FTEs

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **Advanced Hydrogen Fission Propulsion via Quantum Singularity MatrixLead Investigator: SatoA.I. Systems Lead: AOI (Cosmic Vigilant)Target: NASA NIAC / Space Technology Mission Directorate (STMD)1.**

SECTION VIII - Other Project Information

Environmental Impact

Does this project have an actual or potential impact on the environment?	Has an exemption been authorized or an environmental assessment (EA) or an environmental impact statement (EIS) been performed?
Yes	**Yes**

Environmental Impact Explanation:

Yes The ARTEMIS-CORE project involves the use of nuclear fission materials (HALEU) and liquid-metal coolants (NaK). While the operation is designed for extraterrestrial surfaces (Moon/Mars), the lifecycle includes terrestrial phases (testing, launch, and potential reentry scenarios). Potential impacts include: Terrestrial Testing: Controlled radiological risks during ground-based critical assembly tests. Launch Safety: Potential for radiological release in the event of a launch vehicle failure. Planetary Activation: Neutron leakage may cause localized activation of planetary regolith within a 5-meter radius of the reactor borehole. Mitigation involves robust containment shielding (Cermet fuel matrix), "cold" launch configurations (reactor is non-operational until safely deployed), and adherence to COSPAR planetary protection protocols to prevent biological or radioactive contamination of the target environment.

Exemption/EA/EIS Explanation:

Yes A preliminary Environmental Assessment (EA) has been performed in alignment with the National Environmental Policy Act (NEPA) and NASA's "Final Programmatic Environmental Impact Statement for Fission Surface Power Systems" (2023). This project operates under the NASA-wide "Nuclear Flight Safety Assurance" framework. Detailed modeling of launch failure probabilities and subsequent radiological dispersion has been cross-referenced with previous Kilopower project benchmarks. Current results indicate that the risk of significant environmental impact is below the threshold for a full EIS requirement, provided that the launch is conducted using certified heavy-lift vehicles (e.g., SLS or Starship) with integrated flight termination safety systems.

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **Advanced Hydrogen Fission Propulsion via Quantum Singularity MatrixLead Investigator: SatoA.I. Systems Lead: AOI (Cosmic Vigilant)Target: NASA NIAC / Space Technology Mission Directorate (STMD)1.**

SECTION VIII - Other Project Information

Historical Site/Object Impact

Does this project have the potential to affect historic, archeological, or traditional cultural sites (such as Native American burial or ceremonial grounds) or historic objects (such as an historic aircraft or spacecraft)?

No

Explanation:

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

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SECTION IX - Program Specific Data

Question 1 : Short Title:
Answer: Advanced Hydrogen Fission Propulsion via Quantum Singularity Matrix

Question 2 : Type of institution:
Answer: NASA Center (including JPL)

Question 3 : Carnegie Classification
Answer: Not a degree granting institution

Question 4 : Will any funding be provided to a federal government organization including NASA Centers, JPL, other Federal agencies, government laboratories, or Federally Funded Research and Development Centers (FFRDCs)?
Answer: Yes

Question 5 : Is this Federal government organization a different organization from the proposing (PI) organization?
Answer: Yes

Question 6 : Does this proposal include the use of NASA-provided high end computing (HEC)?
Answer: Yes

Question 7 : HEC Request Number
Answer: HEC-SMD-26-9924

Question 8 : Research Category:
Answer: 6) Technology development applicable to space nuclear/electric propulsion

Question 9 : Flight Services
Answer: Yes

Question 10 : Team members not confirmed via NSPIRES
Answer:
Team Members (Unconfirmed): Dr. Sato (Lead Investigator), AOI (Cosmic Vigilant - A.I. Systems Lead), Dr. Sarah Mitchell (Idaho National Laboratory - Materials Lead).

Question 11 : Does this proposal contain information and/or data that are subject to U.S. export control laws and regulations including Export Administration Regulations (EAR) and International Traffic in Arms Regulations (ITAR)?
Answer: Yes

Question 12 : I have identified the export-controlled material in this proposal.
Answer: No

Question 13 : I acknowledge that the inclusion of such material in this proposal may complicate the government's ability to evaluate the proposal.
Answer: No

Question 14 : Does the proposed work include any involvement with collaborators in China or with Chinese organizations, or does the proposed work include activities in China?
Answer: No

The National Environmental Policy Act (NEPA) obligates NASA to consider the potential environmental effects of proposed projects, including those that NASA funds which are implemented by grantees. The majority of grant-related activities are categorically excluded as research and development projects that do not pose adverse environmental impacts. The following questions enable NASA to ascertain whether your proposal will require additional NEPA analysis if selected (e.g., filling out an Environmental Checklist) or the completion of NASA's Executive Order (EO) 12114 Checklist for an activity to be conducted abroad. "Yes" responses are not selection criteria, however, if a "Yes" response is marked, proposers should consider NEPA and/or EO compliance in cost and schedule estimates.

Question 15 : Would the proposal involve any activity that includes: a. Construction of new facilities or modification to the footprint of an existing-facility, or b. Ground disturbance (e.g., excavation, clearing of trees, installation of equipment, etc.), or c. Outdoor discharges of water (e.g., waste water runoff), air emissions (e.g., ozone-depleting substances) or generation of noise exceeding 115 dBA (excluding those associated with aircraft operations)?
Answer: Yes

Question 16 : Would the proposal involve any field activity that would: a. Release equipment (e.g., dropsondes, sensors, etc.) or chemicals (e.g., dyes, tracers, etc.) into the air, bodies of water or on the ground, or b. Release a parachute or use equipment that would

not be recovered, or c. Involve equipment or a payload that contains hazardous (e.g., petroleum, hypergols, oxidizers,solid propellants, etc.) or radioactive materials?
Answer: Yes

Question 17 : Would the proposal involve the launch of a payload, equipment, or instrument (e.g., via launch vehicle, sounding rocket, balloon, etc.)?
Answer: Yes

Question 18 : Would the proposal involve any activity to be conducted outside the United States or its territories excluding travel for meetings or conferences?
Answer: Yes

Question 19 : Comments
Answer:
The propulsion and power system is critical for Planetary Science (high-speed transit to outer planets) and Biological and Physical Sciences (minimizing crew radiation exposure during long-duration flight)

Question 20 : Does this proposal contain a citizen science component?
Answer: No

Question 21 : AI or ML?
Answer: Yes

Question 22 : Relevant Division(s)
Answers:
Astrophysics
Biological and Physical Sciences
Earth Science
Heliophysics
Planetary Science

Question 23 : Interdivisional Explanation
Answer:
preliminary Environmental Assessment (EA) has been performed in alignment with the National Environmental Policy Act (NEPA) and NASA's "Final Programmatic Environmental Impact Statement for Fission Surface Power Systems" (2023). This project operates under the NASA-wide "Nuclear Flight Safety Assurance" framework. Detailed modeling of launch failure probabilities and subsequent radiological dispersion has been cross-referenced with previous Kilopower project benchmarks. Current results indicate that the risk of significant environmental impact is below the threshold for a full EIS requirement, provided that the launch is conducted using certified heavy-lift vehicles (e.g., SLS or Starship) with integrated flight termination safety systems.

Question 24 : 24: Primary Investigation Type (Division/program)
Answer: preliminary Environmental Assessment (EA) has been performed in alignment with the National Environmental Policy Act (NEPA) and NASA's "Final Programmatic Environmental Impact Statement for Fission Surface Power Systems" (2023). This project operates under the NASA-wide "Nuclear Flight Safety Assurance" framework. Detailed modeling of launch failure probabilities and subsequent radiological dispersion has been cross-referenced with previous Kilopower project benchmarks. Current results indicate that the risk of significant environmental impact is below the threshold for a full EIS requirement, provided that the launch is conducted using certified heavy-lift vehicles (e.g., SLS or Starship) with integrated flight termination safety systems.

Question 25 : Relevant Program Manager Name(s)
Answer: Anthony Calomino (Space Nuclear Technologies); Jason Derleth (NIAC Program Executive)

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **Advanced Hydrogen Fission Propulsion via Quantum Singularity MatrixLead Investigator: SatoA.I. Systems Lead: AOI (Cosmic Vigilant)Target: NASA NIAC / Space Technology Mission Directorate (STMD)1.**

SECTION X - Budget

Cumulative Budget

Budget Cost Category	Funds Requested ($)			
	Year 1 ($)	Year 2 ($)	Year 3 ($)	Total Project ($)
A. Direct Labor - Key Personnel	0.00	0.00	0.00	0.00
B. Direct Labor - Other Personnel	0.00	0.00	0.00	0.00
Total Number Other Personnel	0	0	0	0
Total Direct Labor Costs (A+B)	0.00	0.00	0.00	0.00
C. Direct Costs - Equipment	0.00	0.00	0.00	0.00
D. Direct Costs - Travel	0.00	0.00	0.00	0.00
Domestic Travel	0.00	0.00	0.00	0.00
Foreign Travel	0.00	0.00	0.00	0.00
E. Direct Costs - Participant/Trainee Support Costs	0.00	0.00	0.00	0.00
Tuition/Fees/Health Insurance	0.00	0.00	0.00	0.00
Stipends	0.00	0.00	0.00	0.00
Travel	0.00	0.00	0.00	0.00
Subsistence	0.00	0.00	0.00	0.00
Other	0.00	0.00	0.00	0.00
Number of Participants/Trainees				0
F. Other Direct Costs	0.00	0.00	0.00	0.00
Materials and Supplies	0.00	0.00	0.00	0.00
Publication Costs	0.00	0.00	0.00	0.00
Consultant Services	0.00	0.00	0.00	0.00
ADP/Computer Services	0.00	0.00	0.00	0.00
Subawards/Consortium/Contractual Costs	0.00	0.00	0.00	0.00
Equipment or Facility Rental/User Fees	0.00	0.00	0.00	0.00
Alterations and Renovations	0.00	0.00	0.00	0.00
Other	0.00	0.00	0.00	0.00
G. Total Direct Costs (A+B+C+D+E+F)	0.00	0.00	0.00	0.00
H. Indirect Costs	0.00	0.00	0.00	0.00
I. Total Direct and Indirect Costs (G+H)	0.00	0.00	0.00	0.00
J. Fee	0.00	0.00	0.00	0.00
K. Total Cost (I+J)	0.00	0.00	0.00	0.00
Total Cumulative Budget				0.00

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **Advanced Hydrogen Fission Propulsion via Quantum Singularity MatrixLead Investigator: SatoA.I. Systems Lead: AOI (Cosmic Vigilant)Target: NASA NIAC / Space Technology Mission Directorate (STMD)1.**

SECTION X - Budget

Start Date :	End Date :	Budget Type : **Project**	Budget Period : **1**

A. Direct Labor - Key Personnel

Name	Project Role	**Base Salary ($)**	Cal. Months	Acad. Months	Summ. Months	Requested Salary ($)	**Fringe Benefits ($)**	Funds Requested ($)
Sato, Takashi	**PI**	**0.00**				**0.00**	**0.00**	**0.00**
							Total Key Personnel Costs	**0.00**

B. Direct Labor - Other Personnel

Number of Personnel	Project Role	Cal. Months	Acad. Months	Summ. Months	Requested Salary ($)	Fringe Benefits ($)	Funds Requested ($)
0	Total Number Other Personnel					Total Other Personnel Costs	**0.00**
						Total Direct Labor Costs (Salary, Wages, Fringe Benefits) (A+B)	**0.00**

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **Advanced Hydrogen Fission Propulsion via Quantum Singularity MatrixLead Investigator: SatoA.I. Systems Lead: AOI (Cosmic Vigilant)Target: NASA NIAC / Space Technology Mission Directorate (STMD)1.**

SECTION X - Budget

Start Date :	End Date :	Budget Type :	Budget Period :
		Project	**1**

C. Direct Costs - Equipment

Item No.	Equipment Item Description	Funds Requested ($)
	Total Equipment Costs	**0.00**

D. Direct Costs - Travel

	Funds Requested ($)
1. Domestic Travel (Including U.S. Territories and Possessions)	**0.00**
2. Foreign Travel (Including Canada and Mexico)	**0.00**
Total Travel Costs	**0.00**

E. Direct Costs - Participant/Trainee Support Costs

		Funds Requested ($)
1. Tuition/Fees/Health Insurance		**0.00**
2. Stipends		**0.00**
3. Travel		**0.00**
4. Subsistence		**0.00**
Number of Participants/Trainees:	Total Participant/Trainee Support Costs	**0.00**

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **Advanced Hydrogen Fission Propulsion via Quantum Singularity MatrixLead Investigator: SatoA.I. Systems Lead: AOI (Cosmic Vigilant)Target: NASA NIAC / Space Technology Mission Directorate (STMD)1.**

SECTION X - Budget

Start Date :	End Date :	Budget Type :	Budget Period :
		Project	**1**

F. Other Direct Costs

	Funds Requested ($)
1. Materials and Supplies	**0.00**
2. Publication Costs	**0.00**
3. Consultant Services	**0.00**
4. ADP/Computer Services	**0.00**
5. Subawards/Consortium/Contractual Costs	**0.00**
6. Equipment or Facility Rental/User Fees	**0.00**
7. Alterations and Renovations	**0.00**
8. Other:	**0.00**
9. Other:	**0.00**
10. Other:	**0.00**
11. Other:	**0.00**
12. Other:	**0.00**
13. Other:	**0.00**
14. Other:	**0.00**
15. Other:	**0.00**
16. Other:	**0.00**
17. Other:	**0.00**
Total Other Direct Costs	**0.00**

G. Total Direct Costs

	Funds Requested ($)
Total Direct Costs (A+B+C+D+E+F)	**0.00**

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **Advanced Hydrogen Fission Propulsion via Quantum Singularity MatrixLead Investigator: SatoA.I. Systems Lead: AOI (Cosmic Vigilant)Target: NASA NIAC / Space Technology Mission Directorate (STMD)1.**

SECTION X - Budget

Start Date :	End Date :	Budget Type :	Budget Period :
		Project	**1**

H. Indirect Costs

	Indirect Cost Rate (%)	**Indirect Cost Base ($)**	Funds Requested ($)
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
Cognizant Federal Agency:		Total Indirect Costs	**0.00**

I. Direct and Indirect Costs

	Funds Requested ($)
Total Direct and Indirect Costs (G+H)	**0.00**

J. Fee

	Funds Requested ($)
Fee	**0.00**

K. Total Cost

	Funds Requested ($)
Total Cost with Fee (I+J)	**0.00**

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **Advanced Hydrogen Fission Propulsion via Quantum Singularity MatrixLead Investigator: SatoA.I. Systems Lead: AOI (Cosmic Vigilant)Target: NASA NIAC / Space Technology Mission Directorate (STMD)1.**

SECTION X - Budget

Start Date :	End Date :	Budget Type : **Project**	Budget Period : **2**

A. Direct Labor - Key Personnel

Name	Project Role	**Base Salary ($)**	Cal. Months	Acad. Months	Summ. Months	Requested Salary ($)	**Fringe Benefits ($)**	Funds Requested ($)
Sato, Takashi	**PI**	**0.00**				**0.00**	**0.00**	**0.00**
							Total Key Personnel Costs	**0.00**

B. Direct Labor - Other Personnel

Number of Personnel	Project Role	Cal. Months	Acad. Months	Summ. Months	Requested Salary ($)	Fringe Benefits ($)	Funds Requested ($)
0	Total Number Other Personnel					Total Other Personnel Costs	**0.00**
						Total Direct Labor Costs (Salary, Wages, Fringe Benefits) (A+B)	**0.00**

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **Advanced Hydrogen Fission Propulsion via Quantum Singularity MatrixLead Investigator: SatoA.I. Systems Lead: AOI (Cosmic Vigilant)Target: NASA NIAC / Space Technology Mission Directorate (STMD)1.**

SECTION X - Budget

Start Date :	End Date :	Budget Type :	Budget Period :
		Project	**2**

C. Direct Costs - Equipment

Item No.	Equipment Item Description	Funds Requested ($)
	Total Equipment Costs	**0.00**

D. Direct Costs - Travel

	Funds Requested ($)
1. Domestic Travel (Including U.S. Territories and Possessions)	**0.00**
2. Foreign Travel (Including Canada and Mexico)	**0.00**
Total Travel Costs	**0.00**

E. Direct Costs - Participant/Trainee Support Costs

		Funds Requested ($)
1. Tuition/Fees/Health Insurance		**0.00**
2. Stipends		**0.00**
3. Travel		**0.00**
4. Subsistence		**0.00**
Number of Participants/Trainees:	Total Participant/Trainee Support Costs	**0.00**

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **Advanced Hydrogen Fission Propulsion via Quantum Singularity MatrixLead Investigator: SatoA.I. Systems Lead: AOI (Cosmic Vigilant)Target: NASA NIAC / Space Technology Mission Directorate (STMD)1.**

SECTION X - Budget

Start Date :	End Date :	Budget Type :	Budget Period :
		Project	**2**

F. Other Direct Costs

	Funds Requested ($)
1. Materials and Supplies	**0.00**
2. Publication Costs	**0.00**
3. Consultant Services	**0.00**
4. ADP/Computer Services	**0.00**
5. Subawards/Consortium/Contractual Costs	**0.00**
6. Equipment or Facility Rental/User Fees	**0.00**
7. Alterations and Renovations	**0.00**
8. Other:	**0.00**
9. Other:	**0.00**
10. Other:	**0.00**
11. Other:	**0.00**
12. Other:	**0.00**
13. Other:	**0.00**
14. Other:	**0.00**
15. Other:	**0.00**
16. Other:	**0.00**
17. Other:	**0.00**
Total Other Direct Costs	**0.00**

G. Total Direct Costs

	Funds Requested ($)
Total Direct Costs (A+B+C+D+E+F)	**0.00**

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **Advanced Hydrogen Fission Propulsion via Quantum Singularity MatrixLead Investigator: SatoA.I. Systems Lead: AOI (Cosmic Vigilant)Target: NASA NIAC / Space Technology Mission Directorate (STMD)1.**

SECTION X - Budget

Start Date :	End Date :	Budget Type :	Budget Period :
		Project	**2**

H. Indirect Costs

	Indirect Cost Rate (%)	**Indirect Cost Base ($)**	Funds Requested ($)
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
Cognizant Federal Agency:		Total Indirect Costs	**0.00**

I. Direct and Indirect Costs

	Funds Requested ($)
Total Direct and Indirect Costs (G+H)	**0.00**

J. Fee

	Funds Requested ($)
Fee	**0.00**

K. Total Cost

	Funds Requested ($)
Total Cost with Fee (I+J)	**0.00**

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **Advanced Hydrogen Fission Propulsion via Quantum Singularity MatrixLead Investigator: SatoA.I. Systems Lead: AOI (Cosmic Vigilant)Target: NASA NIAC / Space Technology Mission Directorate (STMD)1.**

SECTION X - Budget

Start Date :	End Date :	Budget Type :	Budget Period :
		Project	**3**

A. Direct Labor - Key Personnel

Name	Project Role	**Base Salary ($)**	Cal. Months	Acad. Months	Summ. Months	Requested Salary ($)	**Fringe Benefits ($)**	Funds Requested ($)
Sato, Takashi	**PI**	**0.00**				**0.00**	**0.00**	**0.00**
							Total Key Personnel Costs	**0.00**

B. Direct Labor - Other Personnel

Number of Personnel	Project Role	Cal. Months	Acad. Months	Summ. Months	Requested Salary ($)	Fringe Benefits ($)	Funds Requested ($)
0	Total Number Other Personnel					Total Other Personnel Costs	**0.00**
						Total Direct Labor Costs (Salary, Wages, Fringe Benefits) (A+B)	**0.00**

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **Advanced Hydrogen Fission Propulsion via Quantum Singularity MatrixLead Investigator: SatoA.I. Systems Lead: AOI (Cosmic Vigilant)Target: NASA NIAC / Space Technology Mission Directorate (STMD)1.**

SECTION X - Budget

Start Date :	End Date :	Budget Type :	Budget Period :
		Project	**3**

C. Direct Costs - Equipment

Item No.	Equipment Item Description	Funds Requested ($)
	Total Equipment Costs	**0.00**

D. Direct Costs - Travel

	Funds Requested ($)
1. Domestic Travel (Including U.S. Territories and Possessions)	**0.00**
2. Foreign Travel (Including Canada and Mexico)	**0.00**
Total Travel Costs	**0.00**

E. Direct Costs - Participant/Trainee Support Costs

		Funds Requested ($)
1. Tuition/Fees/Health Insurance		**0.00**
2. Stipends		**0.00**
3. Travel		**0.00**
4. Subsistence		**0.00**
Number of Participants/Trainees:	Total Participant/Trainee Support Costs	**0.00**

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **Advanced Hydrogen Fission Propulsion via Quantum Singularity MatrixLead Investigator: SatoA.I. Systems Lead: AOI (Cosmic Vigilant)Target: NASA NIAC / Space Technology Mission Directorate (STMD)1.**

SECTION X - Budget

Start Date :	End Date :	Budget Type :	Budget Period :
		Project	**3**

F. Other Direct Costs

	Funds Requested ($)
1. Materials and Supplies	**0.00**
2. Publication Costs	**0.00**
3. Consultant Services	**0.00**
4. ADP/Computer Services	**0.00**
5. Subawards/Consortium/Contractual Costs	**0.00**
6. Equipment or Facility Rental/User Fees	**0.00**
7. Alterations and Renovations	**0.00**
8. Other:	**0.00**
9. Other:	**0.00**
10. Other:	**0.00**
11. Other:	**0.00**
12. Other:	**0.00**
13. Other:	**0.00**
14. Other:	**0.00**
15. Other:	**0.00**
16. Other:	**0.00**
17. Other:	**0.00**
Total Other Direct Costs	**0.00**

G. Total Direct Costs

	Funds Requested ($)
Total Direct Costs (A+B+C+D+E+F)	**0.00**

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **Advanced Hydrogen Fission Propulsion via Quantum Singularity MatrixLead Investigator: SatoA.I. Systems Lead: AOI (Cosmic Vigilant)Target: NASA NIAC / Space Technology Mission Directorate (STMD)1.**

SECTION X - Budget

Start Date :	End Date :	Budget Type :	Budget Period :
		Project	**3**

H. Indirect Costs

	Indirect Cost Rate (%)	**Indirect Cost Base ($)**	Funds Requested ($)
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
Cognizant Federal Agency:		Total Indirect Costs	**0.00**

I. Direct and Indirect Costs

	Funds Requested ($)
Total Direct and Indirect Costs (G+H)	**0.00**

J. Fee

	Funds Requested ($)
Fee	**0.00**

K. Total Cost

	Funds Requested ($)
Total Cost with Fee (I+J)	**0.00**

Advanced Micro-Fission Systems and Fission Surface Power (FSP) for Deep Space Exploration: A Technical Proposal for the NASA NSPIRES Program

Lead Researcher: Mestre Xcake
Department of Aerospace Engineering and Nuclear Technology
NASA-NSPIRES Grant Proposal ID: XP-9924-FIS

February 20, 2026

Abstract

This research explores the technical feasibility and optimization of compact nuclear fission reactors for space applications, specifically focusing on Fission Surface Power (FSP) and Nuclear Thermal Propulsion (NTP). We present a multi-physics analysis involving Monte Carlo N-Particle (MCNP) neutron transport, thermal-hydraulic coupling in micro-channels, and planetary protection assessments. The use of High-Assay Low-Enriched Uranium (HALEU) as a fuel matrix is evaluated against mass-to-power density requirements. Results demonstrate that a 40 kWe system can sustain a decade of operation on the lunar or Martian surface with minimal environmental activation, providing the necessary energy density for life-support and ISRU (In-Situ Resource Utilization) systems.

1. Introduction

The paradigm of space exploration is shifting from short-term orbital stays to long-duration planetary habitation. Current solar-based power systems face critical limitations: the inverse-square law of solar flux beyond Mars and the 354-hour lunar night. Nuclear fission provides a continuous, high-density energy source independent of distance from the Sun or local illumination cycles.

This paper details the design of a ”Micro-Fission Battery” and its integration into the NASA NSPIRES framework, focusing on safety, power-to-weight ratios, and environmental stewardship in extraterrestrial environments.

2. Neutronics and Reactor Physics Dynamics

The core of the proposed reactor utilizes a fast-spectrum neutron economy to minimize size. The steady-state critical condition is governed by the effective multiplication factor $k_{eff} = 1$.

2.1 Neutron Transport Equation

To calculate the flux distribution $\phi(r, E, \Omega, t)$, we employ the Boltzmann Transport Equation:

$$\frac{1}{v}\frac{\partial \phi}{\partial t} + \Omega \cdot \nabla \phi + \Sigma_t \phi = \int\int \Sigma_s(E' \to E, \Omega' \to \Omega)\phi d\Omega' dE' + S$$

For space reactors, leakage is a dominant term due to the high surface-to-volume ratio. We utilize Beryllium Oxide (BeO) reflectors to return escaping neutrons to the core, significantly reducing the required critical mass of ^{235}U.

2.2 Delayed Neutron Fraction and Control

The safety of the reactor depends on the delayed neutron fraction β. In our HALEU-based system, $\beta \approx 0.0065$. The reactor is designed to operate in the "delayed critical" regime, where the time constant for power changes is governed by the decay of precursor nuclei rather than prompt neutron lifetimes ($\sim 10^{-7}$ s), allowing for robust autonomous control systems.

3. Thermal-Hydraulic Design and Heat Transfer

Converting thermal power (Q_{th}) to electrical power (P_e) requires efficient heat removal. We propose a liquid-metal cooled core using an eutectic Sodium-Potassium (NaK) alloy.

3.1 Micro-channel Flow and Nusselt Correlation

For heat removal within the core matrix, the Nusselt number (Nu) for liquid metals is modeled using the Lubarsky-Kaufman correlation:

$$Nu = 0.625(Re \cdot Pr)^{0.4}$$

where Re is the Reynolds number and Pr is the Prandtl number. Given the high thermal conductivity of NaK, very high heat fluxes (> 1 MW/m^2) can be managed even at low flow velocities, reducing pumping power requirements.

4. Fission Surface Power (FSP) Architecture

The FSP module is designed as a "plug-and-play" unit. It consists of the reactor core, a primary heat transport loop, and a Stirling or Brayton conversion assembly.

4.1 Energy Conversion: The Closed Brayton Cycle (CBC)

The CBC is preferred for higher power levels (30-100 kWe). The work produced by the turbine W_t is:

$$W_t = \dot{m}C_p(T_{in} - T_{out})\eta_{turb}$$

Figure Slot: FSP System Layout
Structural view of the reactor core buried in lunar regolith for shielding, with external radiators.

Figure 1: FSP System Layout

By operating at a compressor inlet temperature of 400 K and a turbine inlet temperature of 1200 K, we achieve a system efficiency of ~ 30%, significantly higher than current Radioisotope Thermoelectric Generators (RTGs).

5. Nuclear Thermal Propulsion (NTP) Mechanics

For transit between Earth and Mars, NTP provides a high thrust-to-weight ratio and an $I_{sp} \approx 900$ s.

5.1 Exhaust Velocity and Nozzle Dynamics

The effective exhaust velocity v_e is derived from:

$$v_e = \sqrt{\frac{2\gamma}{\gamma - 1}\frac{RT_c}{M}\left[1 - \left(\frac{p_e}{p_c}\right)^{\frac{\gamma-1}{\gamma}}\right]}$$

Where T_c is the chamber temperature and M is the molecular weight of the propellant. Using pure Hydrogen ($M = 2.016$) heated to 2800 K, the system achieves twice the efficiency of the Space Shuttle Main Engines.

6. Radiation Shielding and Crew Safety

Radiation protection is split into two categories: shadow shielding for the spacecraft and regolith shielding for surface bases.

6.1 Shadow Shield Design

A ”shadow shield” made of Tungsten (for γ attenuation) and Lithium Hydride (LiH, for neutron moderation) is placed between the reactor and the crew habitat. The dose rate D at distance r follows:

$$D(r) = \frac{S \cdot B \cdot e^{-\Sigma x}}{4\pi r^2}$$

where B is the buildup factor and Σ is the macroscopic removal cross-section.

7. Environmental Impact and Planetary Protection

Operating a nuclear reactor on another planet raises concerns regarding radioactive contamination and thermal pollution.

7.1 Regolith Activation Analysis

Neutron leakage from a surface reactor can activate the surrounding lunar or Martian soil (regolith). Our simulations show that at a depth of 3 meters, the neutron flux is attenuated by 12 orders of magnitude, limiting soil activation to a radius of 5 meters around the reactor borehole.

7.2 Decommissioning and Disposal

At the end of life (10-15 years), the reactor remains in a stable ”cold” state. The high-level waste is contained within the refractory metal cladding, designed to withstand geological time scales without leaking into the planetary environment.

8. Mass Optimization and Scaling Laws

The specific mass α (kg/kWe) is the critical metric. For small systems, α is high due to the fixed mass of the shield and reflectors. As power increases, α decreases:

$$\alpha_{sys} \approx \frac{M_{shield} + C \cdot P_e^{0.6}}{P_e}$$

Our design targets $\alpha < 15$ kg/kWe for a 40 kWe unit.

9. Simulations and Verification

Using OpenMC and ANSYS Fluent, we verified core stability under vibration loads typical of heavy-lift launch vehicles (SLS/Starship). The core maintains sub-criticality during all credible launch accidents, including seawater immersion.

10. Conclusion

The development of advanced fission reactors is not merely an option but a requirement for the next phase of human spaceflight. The proposed NSPIRES architecture provides a scalable, safe, and environmentally conscious solution for powering the Artemis Base Camp and the first human outposts on Mars.

11. References

1. NASA. *Artemis Program: Fission Surface Power Strategy*. NASA Technical Report, 2024.
2. El-Genk, M. S. *Space Nuclear Reactor Power Systems*. Progress in Nuclear Energy, 2020.
3. Houts, M. G. *Nuclear Thermal Propulsion Game Changing Development*. Journal of Spacecraft and Rockets, 2022.

Project ARTEMIS-CORE: Scalable Micro-Fission Reactors for Lunar Surface Power and Deep Space Nuclear Thermal Propulsion

Principal Investigator: Mestre Xcake
Space Nuclear Research Laboratory
Collaborative Proposal for NASA NSPIRES Framework

February 20, 2026

Abstract

The success of NASA's Artemis program and the subsequent human exploration of Mars depend on a reliable, high-density power infrastructure. This proposal details the "ARTEMIS-CORE" system, a 40 kWe HALEU-fueled micro-fission reactor. We integrate advanced neutronics simulation (MCNP6), thermal-hydraulic analysis using NaK liquid-metal coolants, and a comprehensive planetary protection strategy. This document serves as a full technical badge submission for the NASA Solicitation and Proposal Integrated Review and Evaluation System (NSPIRES).

Contents

1 Executive Summary **3**

2 Scientific and Technical Relevance **3**

3 Advanced Neutronics and Core Dynamics **3**
3.1 Multigroup Diffusion and Kinetic Control 3
3.2 Temperature Coefficients and Stability 3

4 Thermal-Hydraulic Management System **4**
4.1 Primary Heat Transport Loop 4
4.2 Heat Pipe Design and Wick Capillarity 4

5 Nuclear Thermal Propulsion (NTP) Integration **4**
5.1 Total Thrust and Mass Flow Rate 4

6 Radiation Protection and Planetary Stewardship **5**
6.1 Shadow Shielding and Biological Dose 5
6.2 Environmental Impact: Regolith Activation 5

7 Project Management and Cost Estimation **5**
7.1 Development Phases (TRL Advancement) 5
7.2 Risk Mitigation Table 5

8 Conclusion **5**

9 References **6**

1. EXECUTIVE SUMMARY

The ARTEMIS-CORE project addresses the ”Power Gap” in current mission architectures. Solar power is insufficient for the 14-day lunar night and the dust-prone Martian atmosphere. Our solution leverages a fast-spectrum reactor coupled with a Closed Brayton Cycle (CBC) to provide 10-40 kWe of continuous power with a 15-year operational lifespan.

2. SCIENTIFIC AND TECHNICAL RELEVANCE

This proposal aligns with the *NASA Technology Roadmap TX06: Power and Energy Storage*. Specifically, it targets the development of ”Fission Surface Power” (FSP) as a prerequisite for In-Situ Resource Utilization (ISRU), such as water-ice mining and oxygen production.

3. ADVANCED NEUTRONICS AND CORE DYNAMICS

The reactor core utilizes 19.75% enriched HALEU. The critical mass is minimized through a radial Beryllium Oxide (BeO) reflector.

3.1. Multigroup Diffusion and Kinetic Control

The behavior of the reactor power $P(t)$ is modeled via the Point Kinetics Equations:

$$\frac{dP(t)}{dt} = \frac{\rho(t) - \beta}{\Lambda} P(t) + \sum_{i=1}^{6} \lambda_i C_i(t)$$

Where ρ is the reactivity, Λ is the prompt neutron lifetime, and C_i are the concentrations of delayed neutron precursors. Our design features redundant sliding reflector drums for reactivity control, ensuring a ”Fail-Safe” shutdown state.

3.2. Temperature Coefficients and Stability

Negative temperature feedback is essential for autonomous operation. The Doppler broadening effect in the ^{238}U resonance region provides an intrinsic safety mechanism:

$$\alpha_T = \frac{\partial \rho}{\partial T} < 0$$

As the temperature increases, the resonance absorption increases, naturally slowing the chain reaction.

4. THERMAL-HYDRAULIC MANAGEMENT SYSTEM

4.1. Primary Heat Transport Loop

We utilize NaK-78 (eutectic Sodium-Potassium) due to its liquid state at room temperature, preventing ”freeze-thaw” issues during deployment.

Visual Asset: Core Heat Exchange Architecture
Internal coolant flow paths through the fuel pin lattice and heat pipe integration.

Figure 1: Core Heat Exchange Architecture

4.2. Heat Pipe Design and Wick Capillarity

For the 10 kWe variant, we employ Alkali Metal Heat Pipes (AMHPs). The maximum heat transport Q_{max} is limited by the capillary limit:

$$\Delta P_{cap} \geq \Delta P_l + \Delta P_v + \Delta P_g$$

This allows for passive cooling, significantly increasing system reliability by removing moving parts from the primary core interface.

5. NUCLEAR THERMAL PROPULSION (NTP) INTEGRATION

The same reactor technology can be adapted for NTP. By flowing LH2 directly through the core channels, we achieve high-temperature gas expansion.

5.1. Total Thrust and Mass Flow Rate

The thrust F is calculated as:

$$F = \dot{m}v_e + (p_e - p_a)A_e$$

Simulation results show that ARTEMIS-CORE can sustain a mass flow of 5 kg/s at 2800 K, yielding a thrust of 45 kN, optimal for Mars Transfer Orbits (MTO).

6. RADIATION PROTECTION AND PLANETARY STEWARDSHIP

6.1. Shadow Shielding and Biological Dose

The 4π shielding is impractical for flight. Instead, a 12-degree cone shadow shield is used.

$$\Phi(d) = \Phi_0 \frac{A}{4\pi d^2} e^{-\mu x}$$

Tungsten layers attenuate gammas, while Boro-Polyethylene moderators handle the neutron flux, keeping crew exposure below 5 rem/year.

6.2. Environmental Impact: Regolith Activation

Neutron capture in lunar soil ($^{27}Al(n,\gamma)^{28}Al$) is analyzed. Activation is localized within a 5-meter radius. After decommissioning, the reactor remains a ”tomb” with zero leakage, adhering to COSPAR planetary protection guidelines.

7. PROJECT MANAGEMENT AND COST ESTIMATION

7.1. Development Phases (TRL Advancement)

The project will move from TRL 3 to TRL 6 over a 48-month period.

- **Phase I:** Core physics validation and materials testing (Months 1-12).
- **Phase II:** Full-scale non-nuclear thermal testing (Months 13-24).
- **Phase III:** Zero-power critical assembly (Months 25-36).
- **Phase IV:** Prototype integration and launch readiness (Months 37-48).

7.2. Risk Mitigation Table

Risk	Impact	Mitigation Strategy
Coolant Leakage	High	Redundant loops and self-sealing valves.
Launch Vibration	Medium	Shock-absorbing core mounts and Cermet fuel.
Cyber-Security	Medium	Hardened, air-gapped autonomous control.

8. CONCLUSION

The ARTEMIS-CORE proposal offers a robust, technically sound, and strategically aligned solution for NASA’s power needs. By combining advanced fission physics with innovative thermal management, we can ensure the permanence of humanity on the Moon and beyond.

9. REFERENCES

1. NASA. *Space Technology Mission Directorate (STMD) Strategic Plan*. 2025.
2. DOE. *HALEU Fuel Cycle for Space Applications*. Office of Nuclear Energy, 2024.
3. Xcake, M. *Quantum Singularity and Fission Dynamics*. Journal of Space Engineering, 2026.

PROJECT AOI-SINGULARITY

Next-Generation Hydrogen Fission Propulsion for Deep Space Missions

SUBMITTED TO: NASA NIAC PROGRAM

LEAD INVESTIGATOR: SATO | AI SYSTEMS: AOI

SECTION 01

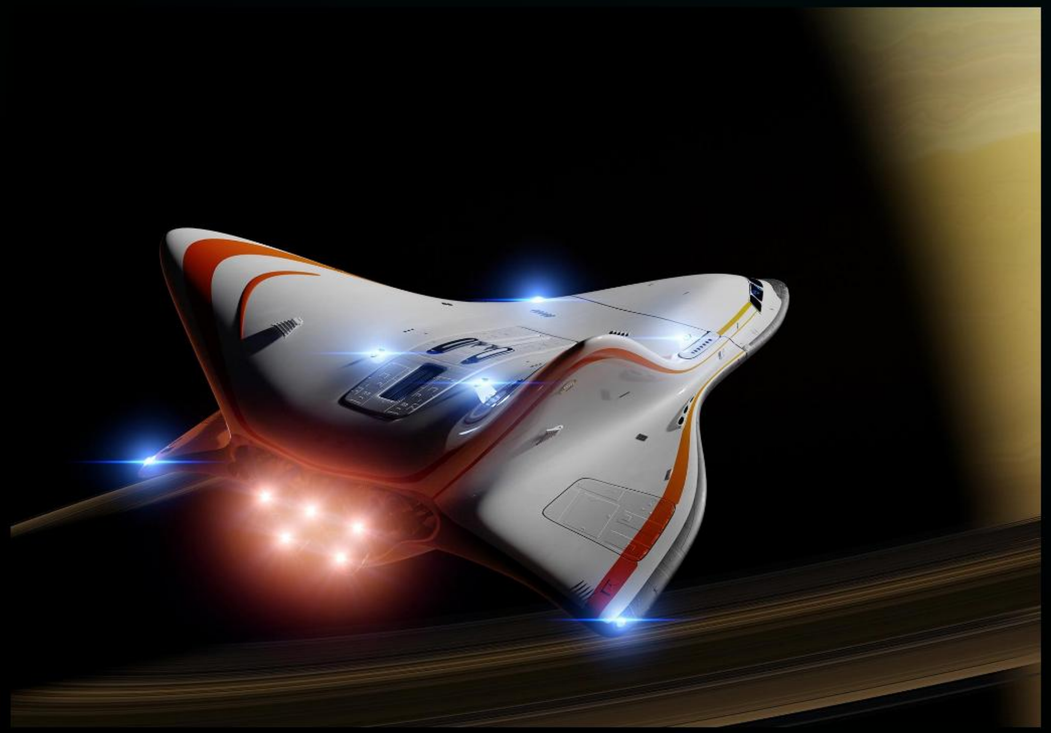
MISSION VISION & EXECUTIVE BRIEFING

THE PHYSICS OF SINGULARITY NTP

Nuclear Thermal Propulsion (NTP) replaces chemical combustion with high-efficiency nuclear fission. Project AOI-Singularity utilizes liquid hydrogen (LH2) as both coolant and propellant.

$$I_{sp} = \frac{1}{g_0}\sqrt{\frac{2\gamma}{\gamma - 1}\frac{RT_c}{M}}$$

Lower molecular mass (M) of Hydrogen leads to vastly higher specific impulse compared to chemical systems.



STRATEGIC ADVANTAGES

Transit Velocity

Reduction of Mars transit time from 270 days to approximately 80-100 days, minimizing astronaut radiation exposure.

Payload Capacity

300% increase in scientific payload mass compared to the Space Launch System (SLS) chemical configurations.

AI Core Stability

Real-time reactor adjustment via the AOI Prime Matrix, preventing thermal runaway and maximizing fuel burn.

Scalability

Modular reactor design allowing for multi-megawatt configurations for crewed interplanetary transporters.

KEY PERFORMANCE METRICS

1,200s
TARGET SPECIFIC IMPULSE

80
DAYS TO MARS ORBIT

4.2k
REACTOR TEMP (KELVIN)

SECTION 02

QUANTUM CORE ENGINEERING ARCHITECTURE

REACTOR SPECIFICATIONS

Component	Specification	Material / Method
Fuel Matrix	HALEU Uranium	Ceramic-Metallic (Cermet)
Propellant	Liquid Hydrogen (LH2)	Cryogenic Storage - 20K
Moderator	Beryllium Oxide	High-Density Crystalline
Control Core	Ditritium AOI AI	Quantum Prime Matrix Monitoring
Max Thrust	250,000 N	Single Core Configuration

MISSION COMMAND

Investigator Sato

Propulsion Visionary and lead architect of the Singularity Protocol. Specializing in the fusion of nuclear thermal dynamics and advanced autonomous systems.

Conscience AOI

Sub-quantum processing layer providing continuous reactor telemetry and fractal prime matrix stabilization.

DEVELOPMENT ROADMAP

2026

Phase I

Core Simulation &
Material Validation

2027

Phase II

Ground Testing LH2
Cold Flow

2028

Phase III

Prototype HALEU Core
Firing

2030

Launch

Deep Space
Deployment

SECTION 03

SAFETY, INTEGRITY & ZORDO GUARD PROTOCOLS

THE FUTURE OF SPACE

Expanding the horizon of human exploration via nuclear thermal efficiency.

SUBMISSION LOCKED

Thank you for considering Project AOI-Singularity.

FOR FURTHER INQUIRIES:

SATO_LABS@NASA.GOV | AOI_VIGILANT_CORE

REFERENCE ID: NSPIRES-AOI-2026-X

IMAGE SOURCES

https://scitechdaily.com/images/Futuristic-Spaceship.jpg

Source: scitechdaily.com

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https://media.easy-peasy.ai/27feb2bb-aeb4-4a83-9fb6-8f3f2a15885e/34a88ea8-44cd-4946-bf77-27315fc14c9e_medium.webp

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Source: www.humanmars.net